# Interrupts (resumed)

## Last time

Exceptional control flow

*Low-level mechanisms*

## Today

Setup/enable interrupts as client

Design of module to manage interrupt system

*Details encapsulated inside module*

*Client interface that is safe, convenient, flexible*

Coordination of activity

*Exceptional and non-exceptional code, dispatch to multiple handlers*

*Data sharing, writing code that can be safely interrupted*

# Looking ahead

This week (last "regular" week) Lab 7, Assign 7

*Polishing touches and crossing the finish line*

*Nab that complete system bonus!*

Brainstorm project ideas and form teams (2 is best)

Labs 8 & 9 will be project team meetings

# Interrupts (so far)

Top-level interrupts system configuration

- External interrupts enabled `mstatus`, `mie` CSRs
- Trap handler address installed in `mtvec` CSR

Transfer of control to enter/exit interrupt code

- Assembly to preserve registers
- Call into C code
- Assembly to restore registers, resume interrupted code

**Today**:

- How to configure system so interrupts are generated
- How to process/clear interrupt

# Three Layers

1. Configure/enable peripheral to generate interrupt on specific event

   E.g. line goes high on certain gpio pin, countdown timer elapsed, char received on uart

2. Configure/enable interrupt source at top-level

3. Globally enable interrupts

*Interrupt generated if and only if all three layers enabled*

*Forgetting to enable one is a common bug*

# HStimer events

## Config timer

- `hstimer_init(hstimer_id, int usecs_interval)`

  Countdown interval (microseconds)

## Enable

- `hstimer_enable(id)` starts countdown, interrupt fires when countdown reaches zero

## Clear interrupt

- `hstimer_interrupt_clear(id)` resets countdown

## References

- Section 3.7 p. 192 in D1-H User Manual
- Review our code in `$CS107E/src/hstimer.c`

# Gpio events

Configure specific event per pin to trigger interrupt

- `gpio_interrupt_config(pin, event, debounce)`

Enable, will generate interrupt on event

- `gpio_interrupt_enable(pin)`

Clear interrupt to reset

- `gpio_interrupt_clear(pin)`

## References

- Section 9.7.3.6 p. 1079 in D1-H User Manual
- Review our code in `$CS107E/src/gpio_interrupt.c`

# Handling event?

Top-level interrupts system configuration

- External interrupts enabled `mstatus`, `mie` CSRs
- Trap handler address installed in `mtvec` CSR

Transfer of control to enter/exit interrupt code

- Assembly to preserve registers
- Call into C code ← Wait, what code is this again?
- Assembly to restore registers, resume interrupted code

# Dispatch to handler

## Each interrupt starts with same actions

- Execute instruction at address stored in `mtvec CSR`
  - `trap_handler` C function
- Single interrupt controller shared by entire program
- How to support different response to timer event vs. button event vs. key event?

## Need handler per-event

- **Function pointers** save the day!
- Each event source has independent handler
- Interrupts module invokes handler registered for source

# Goals for interrupts module

## Convenience, safety

- Abstract away details
- Simple consistent interface
- Defend against mis-use, avoid runtime failures (debugging!)

## Flexible

- Support different use cases (individual handler per interrupt source, independent enable/disable per source)

## Speed

- Minimize number of cycles spent in library
  - Handler may be just a few instructions and runs very often

# Interrupt sources

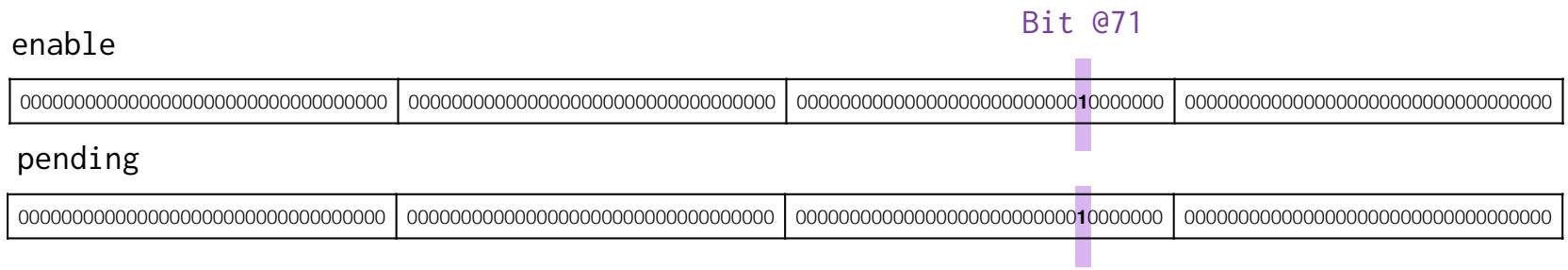
Table 3-9 Interrupt Sources

| Interrupt Number | Interrupt Source |
|---|---|
| 0–15 | Reserved |
| 16 | |
| 17 | |
| 18 | UART0 |
| 19 | UART1 |
| 20 | UART2 |
| 21 | UART3 |
| 22 | UART4 |
| 23 | UART5 |
| 24 | |
| 25 | TWI0 |
| 26 | TWI1 |
| 27 | TWI2 |
| 28 | TWI3 |
| 29 | |
| 75 | TIMER0 |
| 76 | TIMER1 |
| 77 | LRADC |
| 78 | TPADC |
| 79 | WATCHDOG |
| 80 | IOMMU |
| 81 | |
| 82 | VE |
| 83 | |
| 84 | |
| 85 | GPIOB_NS |
| 86 | |
| 87 | GPIOC_NS |
| 88 | |
| 89 | GPIOD_NS |
| 90 | |
| 91 | GPIOE_NS |
| 92 | |

```
enum _interrupt_sources {
    INTERRUPT_SOURCE_UART0 = 18,
    INTERRUPT_SOURCE_UART1 = 19,
    INTERRUPT_SOURCE_UART2 = 20,
    INTERRUPT_SOURCE_UART3 = 21,
    INTERRUPT_SOURCE_UART4 = 22,
    INTERRUPT_SOURCE_UART5 = 23,
    INTERRUPT_SOURCE_HSTIMER0 = 71,
    INTERRUPT_SOURCE_HSTIMER1 = 72,
    INTERRUPT_SOURCE_GPIOB = 85,
    INTERRUPT_SOURCE_GPIOC = 87,
    INTERRUPT_SOURCE_GPIOD = 89,
    INTERRUPT_SOURCE_GPIOE = 91,
    INTERRUPT_SOURCE_GPIOF = 93,
    INTERRUPT_SOURCE_GPIOG = 95,
};
```

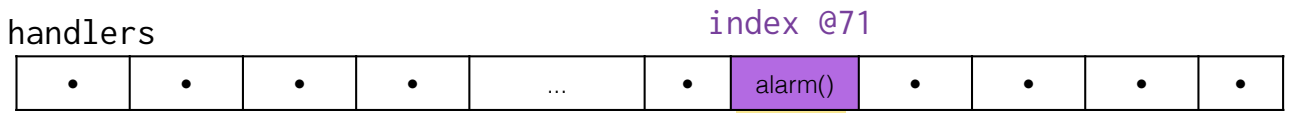
# Implementation

Interrupt peripheral registers, one bit per interrupt source




`INTERRUPT_SOURCE_HSTIMER0 = 71`

Store array of handlers (function pointers)
handler array mirrors structure of peripheral




Top-level trap handler

```
void trap_handler() {
   int idx = index_of_source();
   handlers[idx].fn(pc, handlers[idx].data);
   }
}
```

Client's handler function

```
void alarm(uintptr_t pc, void *data) {
    hstimer_interrupt_clear(HSTIMER0);
    play_sound();
}
```

# Register handler

Client registers handler for a specific interrupt source

- A handler is a function pointer

Array of function pointers, one per interrupt source

- Interrupt source number is index into array

Top-level trap handler dispatches to per-source handler

- Claim register has interrupt source

Aux data can be used to pass information into handler function

- If not needed, aux data can be `NULL`
- Data type is `void *` for flexibility

Review our code in `$CS107E/src/interrupts.c`

# GPIO interrupts

Single interrupt source shared by all GPIO pins within a group

- Need another level of dispatch to support per-pin handler

**`gpio_interrupt_init`** registers handler with top-level **`interrupts`** module

- Shared gpio_interrupt handler receives all gpio events, further dispatches to client's per-pin handler

Internal structure of **`gpio_interrupt`** similar to top-level **`interrupts`**

- Array of handlers, one per pin in group
- Scan event pending register, count zero bits, stop at first set bit, this is index into handler array

Review our code in **`$CS107E/src/gpio_interrupt.c`**

# Client handler function

```
typedef void (*handlerfn_t)(uintptr_t, void *);
```

Two arguments: interrupted **pc**, client data **`void*`**

Ordinary C function (preserve/resume already taken care of)

Handler operation should be lean (other interrupts disabled)

Handler <u>must clear</u> event!

- Otherwise event will continue to refire interrupt forever
- Process for clearing is peripheral-specific

# Interrupt checklist for client

Client must:

- ✅ **Initialize interrupts (and possibly gpio_interrupt)**

Event-specific

- ✅ **Config for desired event**
  - E.g., hstimer countdown reaches zero
- ✅ **Write handler function to process event**
  - Handler acts on event and <u>clears</u> it

- ✅ **Register handler with dispatcher**
  - `gpio_interrupt_register_handler` (if gpio event) or `interrupts_register_handler` (all other sources)
- ✅ **Enable interrupt source**
  - `gpio_interrupt_enable` (if gpio event) or `interrupts_enable_source` (all others)
- ✅ **Globally enable interrupts**
  - Throw the big switch to turn it all on when ready
  - `interrupts_global_enable`

***All steps essential***

*Fiddly code, easy to forget steps, mix up or do in wrong order*

*Typical symptom is absence of action, revisit checklist to find what's missing*

# Sample client code

```
void timer_event(uintptr_t pc, void *aux_data) {
   hstimer_interrupt_clear(HSTIMER0);
   uart_putchar('T');
}

void button_click(uintptr_t pc, void *aux_data) {
   gpio_interrupt_clear(BUTTON);
   uart_putchar('B');
}

void main(void)  {
    uart_init();
    interrupts_init();
    hstimer_init(HSTIMER0, 1000000);
    hstimer_enable(HSTIMER0);
    interrupts_register_handler(INTERRUPT_SOURCE_HSTIMER0, timer_event, NULL);
    interrupts_enable_source(INTERRUPT_SOURCE_HSTIMER0);

    gpio_interrupt_init();
    gpio_interrupt_config(BUTTON, GPIO_INTERRUPT_NEGATIVE_EDGE, true);
    gpio_interrupt_register_handler(BUTTON, button_click, NULL);
    gpio_interrupt_enable(BUTTON);

    interrupts_global_enable();
    while (1) ;
}
```

# code/interrupt_party

# What's left?

An interrupt can fire at any time

- Interrupt handler adds a PS/2 scancode to a queue
- Could do so right as `main` is in middle of removing a scancode from the same queue
- Need to maintain integrity of shared queue

<u>Must write code so that it can be safely interrupted</u>

# Atomicity

main code

```
static int nevents;

    nevents--;
```

interrupt handler

```
static int nevents;

    nevents++;
```

*Q. What is the atomic (i.e., indivisible) unit of computation?*

*Q. Can an update to* `nevents` *be lost when switching between these two code paths?*

# A problem

main code

```
static int nevents;

    nevents--;

li      a4,&nevents
lw      a5,0(a4)
addiw   a5,a5,-1
sw      a5,0(a4)
```

interrupt handler

```
static int nevents;

    nevents++;

li      a4,&nevents
lw      a5,0(a4)
addiw   a5,a5,1
sw      a5,0(a4)
```

How can an increment be lost if interrupt between these two instructions?

# A problem

main code

```
static int nevents;

    nevents--;

li      a4,&nevents
lw      a5,0(a4)
addiw   a5,a5,-1
sw      a5,0(a4)
```

interrupt handler

```
static int nevents;

    nevents++;

li      a4,&nevents
lw      a5,0(a4)
addiw   a5,a5,1
sw      a5,0(a4)
```

Resume instruction uses value previously loaded into **a5**. What happened to increment done by interrupt handler?

# Disabling interrupts

main code

```
interrupts_global_disable();
nevents--;
interrupts_global_enable();
```

interrupt handler

```
nevents++;
```

*Q. Does increment need bracketing also?*

# Preemption and safety

Very hard, lots of bugs.

You'll learn more in CS110/CS140.

Two simple answers

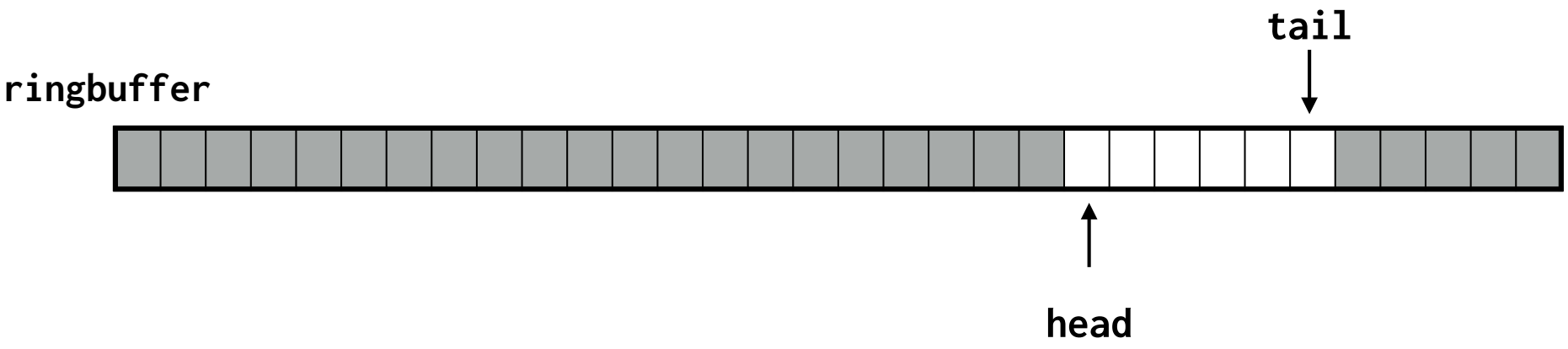
1. Use simple, safe data structures
   - single writer (not always possible)
2. Otherwise, temporarily disable interrupts
   - works if used correctly, easy to get wrong

# Safe ringbuffer

A simple approach to avoid interference is for different code paths to not write to same variables

Queue implemented as ring buffer:

- Enqueue (interrupt) writes element to tail, advances tail
- Dequeue (main) reads element from head, advances head



# Ringbuffer code

```
bool rb_enqueue(rb_t *rb, int elem)
{
    if (rb_full(rb)) return false;

    rb->entries[rb->tail] = elem;
    rb->tail = (rb->tail + 1) % LENGTH; // only changes tail
    return true;
}
```

```
bool rb_dequeue(rb_t *rb, int *p_elem)
{
    if (rb_empty(rb)) return false;

    *p_elem = rb->entries[rb->head];
    rb->head = (rb->head + 1) % LENGTH; // only changes head
    return true;
}
```

Review our code in `$CS107E/src/ringbuffer.c`

# Interrupts in summary

Interrupts allow external events to preempt what's executing and run code immediately

- Needed for responsiveness, e.g., not miss PS/2 scancodes from keyboard when drawing
- Without interrupts, most computers do nothing: they deliver keystrokes, network packets, disk reads, timers, etc.

Simple goal, but config/debug can be tricky!

- Deals with many of the hardest issues in systems

Assign7: update ps2 device driver to read by interrupt